

5-3 Milestone: Tree Data Structure Pseudocode

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CS-300 Analysis and Design

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Function openAndReadFile(filename):

```
try:
    file = open(filename, 'r') // Open file in read mode
    return file.readlines() //Read all lines from the file
except FileNotFoundError:
    print("Error: File not found")
    return []
```

Function parseCourseData(lines):

```
courses = [] //Initialize an empty list to store course objects
for line in lines:
    tokens = line.split(',') //Split line by comma to extract course information
    if len(tokens) < 2: // Check if at least two parameters exist on each line
        print("Error: Insufficient parameters on line:", line)
        continue
    courseNumber = tokens[0]
    title = tokens[1]
    prerequisites = []
    if len(tokens) > 2: //Check if prerequisites are provided
        for i in range(2, len(tokens)):
            prerequisite = tokens[i].strip()
            if not courseExists(prerequisite, courses): //Check if prerequisite exists as a course
                print("Error: Prerequisite", prerequisite, "not found for course", courseNumber)
                continue
            prerequisites.append(prerequisite)
    course = createCourse(courseNumber, title, prerequisites) //Create course object
    courses.append(course) //Store course object in the list
```

```
return courses
```

Function createCourse(courseNumber, title, prerequisites):

```
course = {} //Initialize an empty dictionary for course object
course["courseNumber"] = courseNumber
course["title"] = title
course["prerequisites"] = prerequisites
return course
```

Function courseExists(courseNumber, courses):

```
for course in courses:
    if course["courseNumber"] == courseNumber:
        return True
return False
```

```
filename = "CourseInformation.txt"
```

```
lines = openAndReadFile(filename)
```

```
if lines != []:
```

```
    courses = parseCourseData(lines)
```

```
    for course in courses:
```

```
        print("Course Number:", course["courseNumber"])
```

```
        print("Title:", course["title"])
```

```
        print("Prerequisites:", course["prerequisites"])
```